METHOD 07 / A COMPLETE BEGINNER GUIDE

Partial and residual correlation

How do two FX return series relate after you remove the part explained by a control series? This guide explains residual (partial) correlation in small steps.

You will regress each series on a control factor z, take the leftovers e_x and e_y, and correlate those residuals. That is the idea behind partial correlation in this folder.

Read it in three parts

First: The idea, percent returns, and data rules. **Next:** Symbols, regression formulas, and a hand calculation. **Last:** Majors practice, limits, and software checks.

Pages	What you will learn
2	The idea and whether rho is a percent
3	FX percent math on a worked quote move
4	Which series and control factor to use
5	Research: Basnarkov et al. 2019
6-8	Symbols, formulas, and hand calculation
9-10	Majors context and limits
11-12	Software reproduction and sources

Examples are made-up teaching data. Experiments in this folder have not started. No invented live pair results appear here.

Prepared for the folder: 07 - Partial and Residual Correlation
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02 / THE BASIC IDEA

What is partial / residual correlation?

Sometimes two FX pairs move together mainly because they share a common driver. Partial correlation asks: **after accounting for z, do x and y still move together?**

Two regressions, then one correlation

Fit x = a + b*z + e_x. Fit y = c + d*z + e_y. The residuals e_x and e_y are the parts of x and y not explained by z. Partial correlation is Corr(e_x, e_y).

Is this a percent?

No. Returns may be labelled in percent. The partial correlation rho_partial is on the -1 to +1 scale. It describes linear co-movement of residuals, not a percent of variance explained (that would be R-squared language from regression).

Step	Output
Regress x on z	Residuals e_x
Regress y on z	Residuals e_y
Pearson on e_x, e_y	rho_partial

03 / RETURNS IN PLAIN NUMBERS

FX percent math on one move

From quote **1.1000 to 1.1110**:

```
Simple return = 1.1110 / 1.1000 - 1 = 0.01 (1.00%)
Log return = ln(1.1110 / 1.1000) = about 0.009950 (~0.995%)
```

Partial correlation is usually computed on **return series** (log or simple) aligned in time. The control z_t should be built on the same clock and return definition unless your research note says otherwise.

Control factor examples (conceptual)

z might be a dollar index return, a PCA factor, or another portfolio return. The method does not pick z for you. A bad z choice can leave misleading residual correlation.

04 / DATA AND CONTROL CONSTRUCTION

Which series does this method use?

Returns for x, y, and z

Use matched-period returns for the pair legs and the control. Document demeaning: many workflows use raw returns in regression; others demean first. Stay consistent within an experiment batch.

Why control construction matters

If z is a basket that already contains EUR and GBP, then EURUSD and GBPUSD residuals may look artificially weak or strong because overlap was baked into z. That is **mechanical overlap**, not causal isolation of a unique shock.

Same currencies in basket vs true factor

A well-designed factor targets shared movement you want to remove (e.g. broad USD). A basket copied from the same underlyings without orthogonalisation can remove too much or the wrong part. Write down the economic story before you interpret rho_partial.

Experiments are not started in this folder. Future v001 should name the control series formula and snapshot id explicitly.

05 / RESEARCH NOTES

Where does this fit in FX research?

Basnarkov et al. (2019) [1]

Correlation Patterns in Foreign Exchange Markets. The paper studies structure in FX correlation matrices and motivates tools that separate common components from pair-specific co-movement. Partial and residual correlation are standard ways to ask how pairs relate after conditioning on a factor.

This guide uses their work as the FX research anchor for folder 07. It does not reproduce their empirical tables or claim their results for your snapshot.

Statistical lineage

Partial correlation appears in multivariate statistics textbooks as correlation of residuals from linear adjustment. In FX, the hard part is choosing z, not the final Pearson step on e_x and e_y.

06 / A SMALL MATH DICTIONARY

Meet the symbols

Symbol	Meaning
x_t, y_t	Return series for two pairs (or legs).
z_t	Control factor return at the same time t.
a, b, c, d	Regression intercepts and slopes.
e_x, e_y	Residuals after regressing on z.
rho_partial	Pearson correlation of e_x and e_y.
n	Number of complete time rows.

Each t must have complete x, y, and z. Missing bars drop that row or require an explicit imputation rule recorded in experiment notes.

07 / WHAT EACH PART DOES

The main formulas

```
x_t = a + b * z_t + e_x,t
y_t = c + d * z_t + e_y,t

rho_partial = Corr(e_x, e_y)
```

Step 1: Regress x on z

Ordinary least squares finds a and b that minimise squared errors. e_x,t is the vertical distance from the fitted line at each t.

Step 2: Regress y on z

Do the same separately for y. Use the same z column and the same row set. Do not mix different samples for the two regressions.

Step 3: Correlate residuals

Apply the Pearson formula from folder 01 to e_x and e_y. Sign and strength now refer to co-movement **after** linear removal of z.

Regression details (one control)

```
b = SUM((z_t - z_bar)(x_t - x_bar)) / SUM((z_t - z_bar)^2)
a = x_bar - b * z_bar
e_x,t = x_t - (a + b * z_t)
```

Repeat with y_t to get c, d, and e_y,t. For multiple controls, matrix regression replaces these scalar formulas; this beginner guide stays with one z_t.

Partial vs raw Pearson

If z explains most of x and y, raw Pearson can be high while rho_partial is near zero. The opposite can also happen when z obscures a opposite co-movement. Always report what z was.

09 / WORKED EXAMPLE, PART ONE

Calculate it by hand: start

Five made-up rows. z could stand for a common factor; numbers are not market data.

t	z	x	y
1	1	2	1
2	2	4	3
3	3	5	4
4	4	4	5
5	5	5	6

Regress x on z

```
z_bar = 3, x_bar = 4
b = 6 / 10 = 0.6
a = 4 - 0.6*3 = 2.2
Fitted x_hat = 2.2 + 0.6*z
```

t	x_hat	e_x = x - x_hat
1	2.8	-0.8
2	3.4	0.6
3	4.0	1.0
4	4.6	-0.6
5	5.2	-0.2

Both e_x and e_y will have spread, so Pearson on the residuals is defined.

Calculate it by hand: finish

Regress y on z

```
y_bar = 3.8
b = 6 / 10 = 1.2
c = 3.8 - 1.2*3 = 0.2
Fitted y_hat = 0.2 + 1.2*z
```

t	y_hat	e_y = y - y_hat
1	1.4	-0.4
2	2.6	0.4
3	3.8	0.2
4	5.0	0.0
5	6.2	-0.2

Partial correlation

```
rho_partial = Corr(e_x, e_y) = about 0.8165
```

Positive rho_partial means that after removing linear z effects, the leftover x and y pieces still tend to move in the same direction in this tiny sample. Five rows are far too few for market inference.

11 / EFFECTIVE PRACTICE

Using partial correlation on majors

Typical questions (when experiments exist)

After removing a USD factor, do EURUSD and GBPUSD residuals still correlate? Does a correlation matrix entry reflect EUR-GBP idiosyncrasy or only shared USD beta? Those are design questions, not answers supplied here.

Document z precisely

Write the formula for z_t, its data source, return type, and whether it includes overlapping currencies. Review overlap before calling a residual independence result.

No results in this guide

Folder status is not started. Do not cite pair-specific partial correlations from this PDF. Run v001 with a frozen snapshot and store z construction in NOTES.md.

Limits and common errors

Linear control only

If the true common effect is nonlinear, linear regression leaves structure in residuals. Partial correlation then mixes that leftover structure with genuine pair co-movement.

z is not automatically causal

Removing a statistical factor does not prove counterfactual trading. Overlapping baskets can zero out correlation for mechanical reasons.

Sample size and stability

Many parameters (two regressions plus correlation) on short windows produce unstable rho_partial. Wide confidence intervals are expected unless n is large and the design is stable.

Residual correlation is not partial correlation matrix entry in every package

Some software computes partial correlation via a formula on the full correlation matrix. With one z and OLS residuals, the approaches align for Pearson partial under standard assumptions. Verify your library definition.

Calculate it with software

Python: regress then correlate

```
from math import sqrt
z = [1,2,3,4,5]
x = [2,4,5,4,5]
y = [1,3,4,5,6]
def resid(dep, ctrl):
n = len(dep); mc = sum(ctrl)/n; md = sum(dep)/n
num = sum((ctrl[i]-mc)*(dep[i]-md) for i in range(n))
den = sum((ctrl[i]-mc)**2 for i in range(n))
b = num/den; a = md - b*mc
return [dep[i]-(a+b*ctrl[i]) for i in range(n)]
def pearson(a,b):
n=len(a); ma,mb=sum(a)/n,sum(b)/n
top=sum((a[i]-ma)*(b[i]-mb) for i in range(n))
bot=sqrt(sum((a[i]-ma)**2 for i in range(n))*sum((b[i]-mb)**2 for i in
range(n)))
return top/bot
rho = pearson(resid(x,z), resid(y,z))
print(rho) # about 0.8165
```

statsmodels or sklearn can fit OLS; scipy.stats.pearsonr finishes the last step. Align rows and handle missing data before regression.

Sources and further reading

Links checked on 3 November 2026. Teaching numbers are original. [1] is the FX research anchor for this method folder.

[1] Basnarkov et al. (2019). Correlation Patterns in Foreign Exchange Markets.
FX correlation structure; use for research context and citation in experiment reports.
Open source

[2] SciPy: scipy.stats.pearsonr for the final residual correlation step.
Same Pearson definition as folder 01, applied to e_x and e_y.
Open source

When experiments start, add a third reference row pointing to your frozen snapshot README and the exact z_t formula used in v001.